

Formal Verification of Spiking Neural Networks via Weight-Discretized Quotient Abstractions

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Abstract. Spiking Neural Networks (SNNs) offer biologically plausible, energy-efficient computation but pose severe challenges for formal verification due to state space explosion. Existing quotient model abstractions reduce the state space by partitioning membrane potentials into equivalence classes, but discard synaptic weight information, limiting the properties that can be verified. This paper introduces a *weight-discretized quotient model abstraction* that maps continuous synaptic weights to a compact integer range while preserving the relative contribution of each synapse. The discretization is accompanied by formal correctness guarantees: a Threshold Preservation theorem (completeness) ensures that no fireable configurations are lost, and an Asymptotic Silence theorem (soundness) guarantees that no spurious spikes are introduced. The core biological properties of Leaky Integrate-and-Fire neurons—tonic spiking, integrator behaviour, and excitability—are provably maintained. A topology-dependent scaling analysis derives closed-form state space formulas and shows that the reduction compounds exponentially across neurons: approximately $17\times$ per neuron for a discretization parameter $W = 3$. Empirical validation across seven canonical topologies in the PRISM model checker confirms the theoretical predictions, enabling verification of networks that are otherwise intractable.

Keywords: Spiking Neural Networks · Formal Verification · Model Checking · PRISM · Quotient Abstraction · Weight Discretization

1 Introduction

The evolution of artificial neural networks is conventionally classified into three generations [1]. The first generation, epitomized by McCulloch–Pitts threshold gates, employed binary inputs and outputs and was limited to linearly separable problems. The second generation introduced continuous activation functions and gradient-based optimization via backpropagation, enabling deep learning architectures that have achieved remarkable success in pattern recognition and

sequence modelling. Yet these networks lack the temporal dynamics and energy efficiency of biological neural circuits.

The third generation—Spiking Neural Networks (SNNs)—bridges the gap between machine learning and computational neuroscience. SNNs employ biologically plausible neurons communicating via discrete, asynchronous spikes, where the precise timing of each spike conveys information alongside the aggregate firing rate [1]. This event-driven computation offers substantial power advantages, making SNNs suitable for neuromorphic hardware [2]. However, the non-differentiable Heaviside activation function precludes standard gradient-based optimization, and the stochastic, temporal nature of spike-based computation makes formal verification of correctness significantly harder than for traditional networks. As SNNs are increasingly targeted for safety-critical applications, formal methods providing mathematical guarantees of functional correctness become essential.

This paper presents a *weight-discretized quotient model abstraction* for the formal verification of SNNs via probabilistic model checking. The approach extends the filtration-based quotient model of Baier and Katoen [3] to preserve synaptic weight information during abstraction, enabling verification in the PRISM model checker [4] with drastically reduced state spaces. The contributions are threefold:

1. A **weight discretization scheme** that maps continuous synaptic weights to a finite discrete range while preserving threshold feasibility and relative synaptic contributions (Section 4.1).
2. **Formal correctness proofs:** a *Threshold Preservation* theorem (completeness) and an *Asymptotic Silence* theorem (soundness) guaranteeing that the discretization neither loses fireable configurations nor introduces spurious spikes (Section 4.4).
3. A **topology-dependent state space analysis** with closed-form formulas for DTMC size as a function of network structure, validated empirically across seven canonical topologies (Section 5).

2 Related Work

2.1 Formal Verification of SNNs

De Maria et al. [5] pioneered the modelling of SNNs as timed automata, formalizing Leaky Integrate-and-Fire (LIF) neurons with parameter learning. Their work established key biological properties—tonic spiking, integrator be-

haviour, excitability—as formal verification targets, and introduced the Advice Back-Propagation (ABP) algorithm for supervised parameter inference driven by model-checking counter-examples rather than continuous gradients. This approach was extended to *neuronal archetypes*—primitive micro-circuits such as contralateral inhibition and convergent excitation—allowing macroscopic network properties to be composed from formally verified building blocks [6].

Yao et al. [7] introduced the Refractory-evolve Probabilistic LI&F (RP-LI&F) neuron model, unifying discrete-time refractory dynamics with probabilistic spike generation. Their contract-based verification approach translates SNN topologies into Discrete-Time Markov Chains (DTMCs) and specifies behavioural properties using Probabilistic Computation Tree Logic (PCTL), enabling rigorous assume/guarantee contracts.

These approaches establish the feasibility of formal SNN verification but share a common limitation: the *state space explosion problem*. As network size grows, the DTMC state space grows exponentially in the number of neurons, and existing quotient model abstractions [3] lose synaptic weight information during reduction. The present work addresses both limitations.

2.2 Learning Algorithms for SNNs

Parameter learning in SNNs has motivated diverse approaches. Spike-Timing-Dependent Plasticity (STDP), combined with winner-takes-all lateral inhibition and adaptive thresholds, achieves competitive unsupervised accuracy [8]. Competitive learning frameworks using global inhibition and anti-Hebbian dynamics replace non-local backpropagation with biologically plausible mechanisms [9]. Self-Organizing Maps adapted to the spiking domain use distance-dependent lateral inhibition to create topographically ordered feature maps [10, 11]. For supervised learning at scale, surrogate gradient methods substitute smooth proxy derivatives for the spike function during backpropagation through time [12], while three-factor learning rules with eligibility traces enable online, hardware-compatible training [13].

The present work is complementary to these learning approaches: rather than optimizing SNN parameters, it provides formal guarantees that a given SNN configuration satisfies specified temporal and probabilistic properties.

3 Preliminaries

3.1 Spiking Neural Network Model

An SNN is modelled as a directed graph $G = (V, E)$ where $V = V_{\text{in}} \cup V_{\text{proc}}$ partitions into input neurons and processing neurons, and $E \subseteq V \times V$ represents directed synaptic connections with integer weights $w_e \in [-100, 100]$.

Each processing neuron $n \in V_{\text{proc}}$ follows Leaky Integrate-and-Fire (LIF) dynamics [5, 14]. At each discrete time step, the membrane potential integrates incoming weighted spikes and decays toward rest:

$$p_{n,t+1} = \max \left(0, (1 - \ell) \cdot p_{n,t} + \sum_{i \in \text{In}(n)} w_{i,n} \cdot y_{i,t} \right) \quad (1)$$

where $\ell \in [0, 1]$ is the leak factor. When $p_n \geq P_{\text{rth}}$ (the firing threshold), neuron n emits a spike ($y_n = 1$) and resets to zero.

Firing is *probabilistic*: the potential maps to a discrete threshold level $L \in \{0, \dots, N-1\}$ where N is configurable (1–10), and each level yields a firing probability. Optionally, neurons implement a three-state refractory machine: Normal ($s = 0$), Absolute Refractory Period (ARP, $s = 1$), and Relative Refractory Period (RRP, $s = 2$), with reduced firing probability during RRP scaled by factor α .

3.2 Probabilistic Model Checking

Definition 1. A Discrete-Time Markov Chain (DTMC) is a tuple $\mathcal{D} = (S, s_0, \mathbf{P})$ where S is a finite set of states, $s_0 \in S$ is the initial state, and $\mathbf{P} : S \times S \rightarrow [0, 1]$ is the transition probability matrix satisfying $\sum_{s' \in S} \mathbf{P}(s, s') = 1$ for all $s \in S$ [3].

Behavioural properties are specified in Probabilistic Computation Tree Logic (PCTL), which extends branching-time temporal logic with probabilistic path quantifiers $P_{\bowtie p}[\varphi]$. For instance, $P_{\geq 1}[\mathbf{F}(y_n = 1)]$ asserts that neuron n fires with probability one, while $P_{\geq 1}[\mathbf{G}(y_n = 0)]$ asserts permanent silence.

PRISM [4] is a probabilistic model checker supporting DTMCs. Models are specified as parallel compositions of *modules* with local integer variables and guarded probabilistic transitions, synchronizing via shared labels. The global state space is the Cartesian product of all module state spaces. PRISM offers an *explicit* engine (enumerates reachable states) and a *symbolic BDD* engine (uses

Binary Decision Diagrams via the CUDD library), whose efficiency depends on variable ordering.

3.3 Quotient Model Abstraction

Quotient model abstraction [3, 15] reduces the DTMC state space by partitioning states into equivalence classes. The filtration-based approach maps continuous membrane potentials to a finite class set, preserving probabilistic branching structure. However, the standard filtration treats all synapses uniformly, collapsing weight information: it cannot distinguish strong excitatory from weak excitatory or inhibitory contributions. The weight discretization scheme presented in Section 4 addresses this limitation.

4 Weight-Discretized Quotient Abstraction

The standard filtration-based quotient model abstracts membrane potentials into equivalence classes but treats all synapses uniformly, discarding weight information. This section introduces a *weight discretization scheme* that preserves the relative contribution of each synapse.

4.1 Weight Discretization Function

Definition 2. Given a weight range $[-w_{\max}, w_{\max}]$ (typically $w_{\max} = 100$) and a discretization parameter $W \in \mathbb{N}^+$, the weight discretization function $\delta_W : \mathbb{R} \rightarrow \mathbb{Z}$ is:

$$\delta_W(w) = \left\lfloor w \cdot \frac{W}{w_{\max}} \right\rfloor \quad (2)$$

mapping original weights to the discrete range $[-W, W] \subset \mathbb{Z}$.

The function preserves relative weight magnitudes: for $W = 3$, strong excitatory ($w = 100$) maps to $\delta_3(100) = 3$, medium ($w = 67$) to $\delta_3(67) = 2$, weak ($w = 33$) to $\delta_3(33) = 1$, and inhibitory ($w = -50$) to $\delta_3(-50) = -2$. Weights are stored as static integer constants in the generated PRISM model, contributing no additional state variables.

The *weighted contribution* for neuron n with discretized incoming weights $\{w_1^d, \dots, w_m^d\}$ replaces the binary class evolution of the original quotient model:

$$C_n = \sum_{i=1}^m w_i^d \cdot y_i \quad (3)$$

where $y_i \in \{0, 1\}$ is the spike output of presynaptic neuron i and m is the fan-in.

4.2 Threshold Calibration

Definition 3. *The discretized threshold for a neuron with original threshold T is:*

$$T_d = \left\lceil T \cdot \frac{W}{w_{\max}} \right\rceil \quad (4)$$

The use of ceiling (rather than rounding) ensures $T_d \geq T \cdot W/w_{\max}$, so the discretized neuron is *at least as hard* to fire as the original. This conservative calibration prevents false-positive firings and is essential for the soundness guarantee established in Section .

4.3 Threshold-Dependent Leak Factor

Definition 4. *The discretized leak factor is:*

$$\lambda_d = -\max(1, \lfloor \ell \cdot T_d \rfloor) \quad (5)$$

where $\ell \in [0, 1]$ is the leak rate ($\ell = 1 - r$ with retention rate r).

This formulation links the leak to T_d rather than to the number of equivalence classes k . Since T_d scales with the actual potential range, the decay remains proportional to realistic membrane dynamics. The $\max(1, \dots)$ floor ensures a minimum decay of one unit per step, which is required by the Soundness theorem (Section). The class evolution becomes:

$$c'_n = \text{clamp}(c_n + \Delta(C_n) + \lambda_d, 0, k) \quad (6)$$

where $\Delta(C_n) = \text{clamp}(\lfloor C_n/\gamma \rfloor, -k, k)$ is the class delta function with class width $\gamma = T_d/k$.

4.4 Formal Proofs

Threshold Preservation (Completeness).

Theorem 5. *Let $\mathcal{N}$ be a neuron with weights $\{w_1, \dots, w_m\}$ and threshold T . If $\mathcal{N}$ can fire in a single step (i.e., $\exists \mathbf{y} \in \{0, 1\}^m$ such that $\sum_{i=1}^m w_i \cdot y_i \geq T$), then the discretized neuron $\mathcal{N}'$ with weights $\{\delta_W(w_1), \dots, \delta_W(w_m)\}$ and threshold T_d can also fire.*

Proof. Let $\mathbf{y}^*$ be a firing input pattern with weighted sum $S = \sum w_i y_i^* \geq T$. By the rounding bound, $\delta_W(w_i) \geq w_i \cdot W/w_{\max} - 1/2$, so the discretized sum satisfies:

$$S_d = \sum \delta_W(w_i) \cdot y_i^* \geq S \cdot W/w_{\max} - m^*/2 \quad (7)$$

where $m^* = \sum y_i^* \leq m$ is the number of active inputs. Since $T_d = \lceil T \cdot W/w_{\max} \rceil \leq T \cdot W/w_{\max} + 1$, the sufficient condition for firing is $(S - T) \cdot W/w_{\max} \geq m/2 + 1$. For $W \geq w_{\max} \cdot (m/2 + 1)/T$, the condition is satisfied whenever $S \geq T$. With typical parameters ($W = 3$, $w_{\max} = 100$, $m \leq 10$, $T = 100$), the bound requires $W \geq 6$. $\square$

Remark 6. The cumulative rounding error of $-m/2$ implies that high fan-in neurons require finer discretization. For neurons with $m > 2W$, the rounding noise may exceed the smallest synaptic weight, and a threshold correction factor $T'_d = T_d - \lfloor m/(2W) \rfloor$ can optionally be applied.

Asymptotic Silence (Soundness).

Theorem 7. *Let $\mathcal{N}'$ be a discretized neuron with potential $P_t < T_d$ and leak factor $\lambda_d \leq -1$. If the input is zero for all $t' \geq t$ (i.e., $S_d = 0$ henceforth), then $\mathcal{N}'$ will never fire.*

Proof. Without input, $P_{t+1} = \max(0, P_t + \lambda_d)$. Since $\lambda_d \leq -1$ and $P_t > 0$, the potential strictly decreases: $P_{t+1} \leq P_t - 1$. The sequence converges to the absorbing state $P = 0$. Since the trajectory is non-increasing and starts below T_d , the firing condition $P \geq T_d$ is never met. At $P = 0$, the threshold level is $L = 0$, which maps to firing probability zero, ensuring permanent silence. $\square$

Biological Property Preservation.

The discretized model preserves the core LIF properties formalized by De Maria et al. [5]:

- **Tonic spiking.** Under constant input C_{in} , the neuron has non-zero firing probability iff $C_{\text{in}} > |\lambda_d|$. When satisfied, the potential accumulates to a level

$L > 0$, yielding periodic probabilistic spiking with expected inter-spike interval $\text{ISI} = \lceil T_d / (C_{\text{in}} - |\lambda_d|) \rceil$.

- **Integrator.** The probability of immediate firing on simultaneous inputs reaches 1.0 iff $\sum \delta_{W(w_i)} \geq T_d$. Below threshold, the response is graded via the threshold level mapping.
- **Excitability.** The expected inter-spike interval decreases monotonically as input strength increases, since $\text{ISI} = \lceil T_d / (C_{\text{in}} - |\lambda_d|) \rceil$ is a decreasing function of C_{in} .

5 Topology-Dependent Scaling Limits

To characterize the practical limits of SNN verification, this section derives closed-form formulas for the DTMC state space as a function of network topology and model configuration.

5.1 PRISM Module Decomposition

The PRISM model for an SNN $G = (V, E)$ consists of four module types:

- **GlobalClock:** a step counter with $T_{\text{max}} + 1$ states.
- **Inputs:** $2^{|V_{\text{in}}|}$ states (one binary variable per input neuron).
- **Neuron M_n :** per-neuron state depends on configuration:
 - *Fast* ($k = 10$ threshold levels, no refractory): $|S_n^{\text{fast}}| = 2 \cdot R_n$ where $R_n = P_{\text{max},n} - P_{\text{min},n} + 1$.
 - *Full* ($k = 10$, ARP=2, RRP=4): $|S_n^{\text{full}}| = 3 \cdot 3 \cdot 5 \cdot 2 \cdot R_n = 90 \cdot R_n$.
 - *Discretized* ($W = 3$): $|S_n^{\text{disc}}| = 2 \cdot (P_{\text{max},n}^d + 1)$ where $P_{\text{max},n}^d = T_d + E_n^d$.
- **Transfer $T_{i,j}$:** 2 states per internal edge.

Theorem 8. (State Space Product). *The theoretical state space is the Cartesian product of all module state spaces:*

$$|S_{\text{theory}}| = (T_{\text{max}} + 1) \cdot 2^{|V_{\text{in}}|} \cdot \prod_{n \in V_{\text{proc}}} f_n(C) \cdot 2^{|E_{\text{int}}|} \quad (8)$$

where $f_n(C) = |S_n|$ depends on the model configuration C .

For a chain of N neurons with $P_{\text{rth}} = 100$ and weight $w = 80$: the per-neuron factor is $f_n = 2 \cdot 121 = 242$ (fast precise), dropping to $f_n = 2 \cdot 7 = 14$ (discretized $W = 3$). The per-neuron reduction factor is $242/14 \approx 17.3 \times$.

Theorem 9. (Exponential Reduction). *For a chain of N neurons, the state space ratio between precise and discretized models compounds exponentially:*

$$|S^{\text{precise}}| \Big| S^{\text{disc}} = \prod_{n=1}^N \frac{R_n}{P_{\max,n}^d + 1} \approx 17.3^N \quad (9)$$

For $N = 4$: $17.3^4 \approx 89,500 \times$.

5.2 Empirical Validation

63 PRISM models were generated across 7 canonical topologies (single, chain-2 through chain-4, fork, diamond, convergent), 3 configurations (deterministic, fast, full), and 3 model types (precise, disc. $W = 2$, disc. $W = 3$). Fig. 1 reports the reachable state counts from DTMC exports.

Fig. 1. Reachable DTMC states $|S_{\text{reachable}}|$ by topology and configuration. All precise models use $k = 10$ threshold levels. “OOM” denotes CUDD BDD out-of-memory (2 GB limit).

Topology	Det.	Fast	Full	Disc. $W=3$
Single (1N)	5	6	18	6
Chain-2 (2N)	12	23	246	11
Chain-3 (3N)	23	275	13,180	28
Chain-4 (4N)	42	6,917	362,289	88
Fork (3N)	12	83	3,388	15
Diamond (5N)	15	1,511	OOM	98
Convergent (1N)	17	20	65	16

The most striking pattern is the exponential growth across chain lengths in the precise configurations: from 18 states (single, full) to 362,289 (chain-4, full)—a $20\{, \}000 \times$ increase for just three additional neurons. The discretized column remains remarkably stable, growing from 6 to 88 across the same range.

The per-neuron multiplicative factor for the fast precise model is:

$$r_{\text{fast}} \approx 23/6 \approx 3.8, \quad 275/23 \approx 12.0, \quad 6917/275 \approx 25.2 \quad (10)$$

This acceleration occurs because downstream neurons receive progressively richer input distributions, approaching the theoretical maximum of 484 per neuron as the reachability density ρ increases.

Three further observations emerge:

1. **Fan-out is more expensive than fan-in.** The fork topology (3 neurons, fan-out 2) produces 275 states (fast), while the convergent topology (1 processing neuron, fan-in 2) produces only 20. Parallel branches multiply the module count, whereas convergent inputs only increase the potential range.

2. **BDD memory is the practical bottleneck.** The diamond topology (5 neurons) causes CUDD out-of-memory in the full precise configuration, while the explicit engine succeeds for chain-4 (362,289 states). BDD variable ordering for multi-path topologies creates exponential intermediate representations.
3. **Discretization scales gracefully.** The disc. $W = 3$ diamond completes with 98 states, compared to OOM for the precise full model. Fig. 2 summarizes the estimated maximum network sizes.

Fig. 2. Estimated maximum verifiable network sizes (2 GB CUDD limit).

Configuration	Chain	Diamond	Notes
Det. precise	>10N	>10N	Minimal state space
Fast precise	5–6N	4N	Per-neuron narrowing
Full precise	4N	<4N	ARP/RRP dominate
Fast disc. $W=3$	>10N	7–8N	17× reduction/neuron
Full disc. $W=3$	8N	5–6N	Best precision/tractability

6 Conclusion

This paper presented a weight-discretized quotient model abstraction for the formal verification of spiking neural networks. The discretization function δ_W maps continuous synaptic weights to a compact integer range while preserving threshold feasibility (Theorem 1) and preventing spurious spikes (Theorem 2). The core biological properties of LIF neurons—tonic spiking, integrator behaviour, and excitability [5] — are maintained by the threshold-dependent leak factor λ_d .

The topology-dependent scaling analysis demonstrates that the state space reduction compounds exponentially across neurons: approximately $17 \times$ per neuron for $W = 3$, enabling verification of networks that are otherwise intractable. Empirical validation across seven canonical topologies confirms the theoretical predictions and identifies BDD memory as the binding practical constraint.

Several directions for future work emerge. First, *compositional verification* could exploit the modular structure of neuronal archetypes [6] to verify large networks by composing archetype-level guarantees. Second, *automated W selection* based on fan-in analysis could optimize the precision/tractability trade-off per neuron. Third, extending the framework to *recurrent topologies* would broaden its applicability. Finally, integrating the formal verification loop into biologically plausible learning frameworks [13] — using verified safety constraints

as modulatory third factors—could enable training of SNNs that are intrinsically correct by construction.

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